

# An Analysis of a Primitive Closed Agrarian Economy

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## Abstract

This paper develops a stylized model of a primitive closed agrarian economy composed of two broad social classes: princes and commoners. The framework studies how food production, famine risk, political hierarchy, and investment in stability interact over time. Unlike a purely static allocation model, the present framework introduces endogenous famine probabilities, dynamic elite formation, and nonlinear feedback effects between political power and economic security. The model is intended not as a literal historical reconstruction, but as an analytical abstraction capable of explaining why some agrarian systems stabilize while others collapse into recurrent famine, elite over-expansion, or political fragmentation.

The paper ends with “The End”

## 1 Introduction

Primitive agrarian societies are fundamentally constrained by food production and food storage. Before industrialization, the overwhelming majority of output consisted of agricultural goods, and the survival of the population depended directly on climatic conditions, harvest quality, and political organization. In such systems, elites often emerged as coordinators of military protection, irrigation, storage, taxation, and redistribution. However, elites also consumed surplus resources, creating a tension between productive stabilization and extractive domination.

This paper formalizes that tension mathematically. The model distinguishes between commoners, who receive relatively stable but low consumption, and princes, who capture a larger but riskier surplus. Investment in state capacity reduces famine risk, but maintaining elite structures also introduces the possibility of over-expansion and instability.

The objective is to derive a minimal but dynamically rich system capable of generating:

- stable agrarian hierarchies,
- recurrent famine cycles,
- elite overgrowth,
- collapse equilibria, and
- oscillatory political-economic dynamics.

## 2 Model Structure

Let total population be denoted by:

$$P$$

The population is divided into:

$$p \quad \text{princes}$$

and

$$P - p \quad \text{commoners}$$

The distinction is political as well as economic. Princes represent the administrative, military, and ruling class, whereas commoners represent the productive agricultural base.

The economy produces two possible food outcomes:

$$F > f > 0$$

where:

- $F$  denotes a high-food or surplus state,
- $f$  denotes a low-food or subsistence state.

Similarly, the economy distinguishes between two meal sizes:

$$M > m > 0$$

where:

- $M$  is the consumption level associated with elites,
- $m$  is the consumption level associated with commoners.

For convenience, normalize:

$$M + m = 1$$

This normalization simplifies comparative analysis without changing the structure of the model. Food state probabilities are defined by:

$$p_F = \frac{1}{2} + \theta, \quad p_f = \frac{1}{2} - \theta, \quad \theta \in (0, \frac{1}{2})$$

The parameter  $\theta$  measures the structural productivity advantage of the agrarian system. Higher values imply that surplus years are more likely than subsistence years.

### 3 Food Risk and Allocation

Food allocation is intentionally asymmetric. Commoners receive stable but low consumption, while princes are exposed to higher upside and greater systemic risk.

Define food risk exposure:

$$r \in [0, 1]$$

and a probability mapping:

$$\alpha(r), \quad \alpha'(r) > 0$$

The function  $\alpha(r)$  captures the probability that risky surplus extraction succeeds. Increasing exposure to risk increases the probability of accessing high-surplus outcomes, but also increases vulnerability to famine shocks.

For commoners:

$$r_c = 0, \quad x_c = f$$

Commoners therefore consume a subsistence-level food quantity regardless of the upside state. This assumption reflects the idea that peasants are often insulated from gains but not from scarcity.

Princes instead receive:

$$E[x_\pi] = f + (1 - \lambda)\alpha(r)(F - f)$$

This equation is central to the political economy of the model. Princes receive the baseline subsistence quantity  $f$ , plus an additional surplus term. That surplus depends on:

1. the absence of famine  $(1 - \lambda)$ ,
2. successful risk extraction  $\alpha(r)$ ,
3. the size of the agricultural surplus  $(F - f)$ .

Thus elite prosperity depends simultaneously on political order, environmental stability, and successful extraction.

## 4 Famine Shock Process

Agrarian societies are vulnerable to exogenous shocks such as droughts, floods, crop disease, and warfare disruption. To model this, define a famine shock variable:

$$s \in \{0, 1\}$$

with:

$$\mathbb{P}(s = 1) = \lambda$$

where:

- $s = 1$  denotes famine,
- $\lambda$  is the famine probability.

Food output becomes:

$$F_s = \begin{cases} F, & s = 0 \\ f, & s = 1 \end{cases}$$

In famine states, the economy collapses from surplus production to subsistence output. Expected elite consumption becomes:

$$E[x_\pi] = f + (1 - \lambda)\alpha(F - f)$$

A rise in famine probability directly compresses elite consumption by destroying the surplus component of production.

## 5 Investment and Endogenous Stability

Primitive states often invest in irrigation, granaries, military protection, roads, and food storage systems. These investments reduce the likelihood of catastrophic collapse.

Define aggregate investment:

$$I = pi$$

where:

- $p$  is the number of princes,
- $i$  is investment per prince.

Famine probability is assumed to satisfy:

$$\lambda = \frac{1}{1 + I}$$

This functional form captures diminishing returns to stabilization investment. Initial investments strongly reduce famine risk, but progressively larger expenditures generate smaller marginal improvements.

Differentiating:

$$\frac{d\lambda}{di} < 0, \quad \frac{d\lambda}{dp} < 0$$

Therefore:

- more investment lowers famine risk,
- larger administrative capacity lowers famine risk,
- but maintaining elites may also create long-run extraction pressures.

## 6 Dynamic System

The economy evolves dynamically through interactions between elite size, investment, and famine probability.

### 6.1 Elite Dynamics

Elite growth is governed by:

$$\dot{p} = p[a(1 - \lambda)\alpha - b(G - 1)]$$

The first term represents surplus-driven elite expansion. When famine risk is low and extraction succeeds, elite structures grow.

The second term captures the destabilizing effects of inequality. Excessive inequality generates social strain, administrative inefficiency, succession conflict, and political fragmentation.

Parameters:

- $a$  measures responsiveness of elite expansion,
- $b$  measures instability generated by inequality.

### 6.2 Investment Dynamics

Investment evolves according to:

$$\dot{i} = \gamma \left[ \frac{p\alpha(1 - \lambda)(F - f)}{(1 + pi)^2} - ci \right]$$

The numerator captures surplus available for investment. The denominator introduces diminishing returns and bureaucratic inefficiency.

The parameter:

$$c$$

represents investment cost or depreciation.

The parameter:

$$\gamma$$

controls adjustment speed.

### 6.3 Famine Dynamics

Famine risk evolves according to:

$$\dot{\lambda} = -\beta\lambda^2 pi$$

The quadratic term implies that high famine risk can only be reduced through sufficiently large investment efforts.

Parameter:

$$\beta$$

measures the effectiveness of institutional stabilization.

## 7 Inequality Measure

Define inequality:

$$G = \frac{U_\pi}{U_c}$$

where:

$$U_\pi = M[f + (1 - \lambda)\alpha(F - f)]$$

and:

$$U_c = p_f m$$

The model therefore measures inequality as the ratio of elite welfare to commoner welfare.

A rise in:

$$F - f$$

increases inequality because elites disproportionately capture surplus gains.

A rise in famine risk compresses inequality mechanically by destroying elite surplus.

## 8 Equilibrium Conditions

A steady state satisfies:

$$\dot{p} = \dot{i} = \dot{\lambda} = 0$$

An interior equilibrium requires:

$$pi > 0, \quad \lambda > 0$$

At equilibrium:

- elite expansion balances instability,
- investment balances depreciation,
- stabilization exactly offsets famine persistence.

The equilibrium structure may be stable or unstable depending on parameter values.

## 9 Jacobian Matrix

Local dynamics near equilibrium are characterized by the Jacobian:

$$J = \begin{bmatrix} \frac{\partial \dot{p}}{\partial p} & \frac{\partial \dot{p}}{\partial i} & \frac{\partial \dot{p}}{\partial \lambda} \\ \frac{\partial \dot{i}}{\partial p} & \frac{\partial \dot{i}}{\partial i} & \frac{\partial \dot{i}}{\partial \lambda} \\ \frac{\partial \dot{\lambda}}{\partial p} & \frac{\partial \dot{\lambda}}{\partial i} & \frac{\partial \dot{\lambda}}{\partial \lambda} \end{bmatrix}$$

The signs of the derivatives provide economic intuition:

$$\frac{\partial \dot{\lambda}}{\partial p} < 0, \quad \frac{\partial \dot{\lambda}}{\partial i} < 0$$

meaning that greater administrative scale and investment reduce famine persistence.

The eigenvalues of the Jacobian determine whether the economy converges toward equilibrium, diverges away from it, or oscillates between crises and recovery phases.

## 10 Stability Analysis

### 10.1 Stable Node

If:

$$b(G - 1) \gg a(1 - \lambda)\alpha$$

then instability pressures dominate elite expansion. The system converges toward a stable hierarchy. All eigenvalues satisfy:

$$\Re(\lambda_i) < 0$$

Economically, this corresponds to a disciplined agrarian state with controlled inequality and sufficient stabilization capacity.

### 10.2 Saddle Point

If one eigenvalue becomes positive:

$$\lambda_1 > 0$$

then small disturbances grow over time.

This regime corresponds to political fragility. Minor shocks can trigger famine spirals, elite fragmentation, or collapse.

### 10.3 Oscillatory Regime

If:

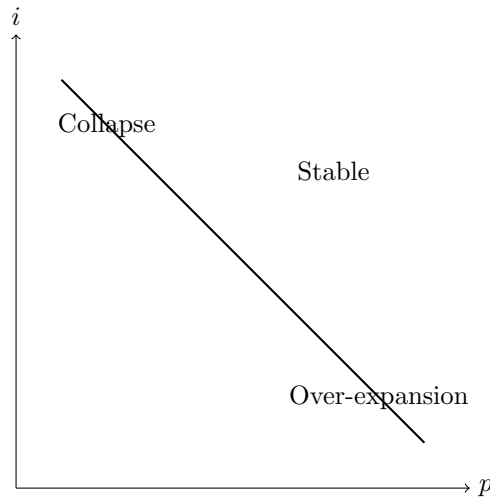
$$\lambda_{1,2} = \alpha \pm i\omega, \quad \alpha \approx 0$$

the system exhibits cyclical behavior similar to a Hopf bifurcation.

Historically, such dynamics resemble recurring agrarian boom-bust cycles:

1. surplus accumulation,
2. elite expansion,
3. rising extraction,
4. instability,
5. collapse,
6. recovery.

## 11 Phase Diagram



The phase diagram summarizes the qualitative structure of the economy:

- low investment and weak elite organization lead to collapse,
- balanced investment and hierarchy generate stability,
- excessive elite growth generates instability and cyclical crisis.

## 12 Regime Classification

| Regime              | Characteristics          | Outcome            |
|---------------------|--------------------------|--------------------|
| Fragmented Collapse | High $\lambda$ , low $i$ | Famine dominance   |
| Stable Hierarchy    | Balanced feedback        | Sustained surplus  |
| Over-expansion      | High $p$                 | Instability cycles |

The table emphasizes that agrarian systems are not characterized by a single equilibrium. Instead, different institutional and environmental conditions generate qualitatively different political-economic outcomes.

## 13 Algorithmic Representation

### Algorithm 1: System Evolution

1. Initialize  $p, i, \lambda$
2. Compute surplus and expected elite utility
3. Update investment dynamics
4. Update famine probability
5. Update elite size
6. Repeat until convergence or instability

This algorithmic interpretation allows the model to be implemented computationally through simulation methods.

## 14 Conclusion

This paper develops a compact but flexible framework for analyzing primitive agrarian economies under endogenous famine risk and political hierarchy. The model demonstrates that elite organization can simultaneously stabilize and destabilize economic systems. Moderate investment reduces famine risk and sustains surplus production, but excessive elite expansion can generate instability and cyclical collapse.

The broader implication is that political hierarchy in pre-industrial societies should not be interpreted purely as extraction. Elite structures may emerge partly as adaptive responses to systemic risk. However, once established, those same structures may evolve toward over-expansion, inequality, and institutional fragility.

The framework therefore provides a bridge between political economy, historical state formation, famine theory, and nonlinear dynamical systems analysis.

**The End**